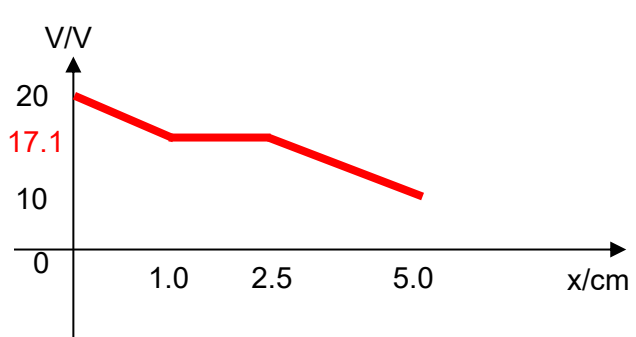
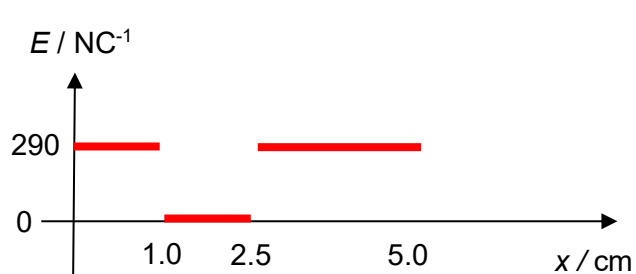


3	
(a)	Electric potential V at a point is defined as the <u>work done per unit charge</u> by an <u>external agent</u> in bringing a <u>small positive</u> point test <u>charge from infinity to that point, without acceleration</u> .
(b)(i)	
	$E = \frac{\Delta V}{\Delta d} = \frac{20 - 10}{3.5} = 2.86 = 2.9 \text{ V cm}^{-1} (\text{NC}^{-1})$ Decrease in V per $\text{cm} = 2.9 \text{ V}$ Potential at horizontal line $= 20 - 2.9 = 17.1 \text{ V}$
	Gradient is the same for both lines.
(b)(ii)	
	Taking to the right as positive. E is negative potential gradient ($E = -\frac{dV}{dr}$)
	$E = 2.9 \text{ V cm}^{-1} = 290 \text{ V m}^{-1}$
(c)(i)	$V = \frac{1.5 \times 10^{-11}}{4\pi\epsilon_0(2.5 \times 10^{-2})}$ $V = 5.395 \text{ V}$
	$I = \frac{5.395}{5.0}$ $I = 1.079 \text{ A}$
	$= 1.08 \text{ A (3sf)}$
(c)(ii)	$I = \frac{V}{R}$ Since I and R are constants, <u>V must be constant.</u>
	$V = \frac{Q}{4\pi\epsilon_0 r}$ $r \propto Q$ Since Q is decreasing, r also decreases to maintain constant V.

(c)(iii)	$V = \frac{Q}{4\pi\epsilon_0 r}$ $r = \frac{Q}{4\pi\epsilon_0 V}$ $r = 1.67 \times 10^9 Q$ $\frac{dr}{dt} = 1.67 \times 10^9 \frac{dQ}{dt}$ Hence $R_{radius} = 1.7 \times 10^9 I$
(c)(iv)	$t = \frac{Q}{I}$ $t = \frac{1.5 \times 10^{-11}}{1.08}$ $t = 1.39 \times 10^{-11} \text{ s (3sf)}$
(c)(v)	$R_{radius} = 1.7 \times 10^9 I$ $\frac{dr}{dt} = 1.7 \times 10^9 I$ $dt = \frac{dr}{1.7 \times 10^9 I}$ $t_r = \frac{(2.5 - 0.5) \times 10^{-2}}{(1.7 \times 10^9) 1.08}$ $t_r = 1.09 \times 10^{-11} \text{ s (3sf)}$
	Since $1.09 \times 10^{-11} \text{ s}$ is smaller than $1.39 \times 10^{-11} \text{ s}$, the balloon will be fully deflated first.
4	